Outils Mathématiques – Calcul Intégral

Intégrales Curvilignes

IC1

$I(f) = \int_{\widehat{AB}} f(M)\,ds \;\rightarrow\; f(M) = \dfrac{1}{x-y}$, et $\Gamma : \; y = \dfrac{x}{2} - 2$ sur $\begin{cases} A(0,-2) \\ B(4,0) \end{cases}$

- $ds = \sqrt{dx^2+dy^2} = \sqrt{1+\left(\dfrac{dy}{dx}\right)^2}\,dx$

  $\dfrac{dy}{dx} = \dfrac{1}{2} \Rightarrow ds = \sqrt{1+\left(\dfrac{1}{2}\right)^2}\,dx = \dfrac{\sqrt{5}}{2}\,dx$

- $f(M) = \dfrac{1}{x-y} = \dfrac{1}{x-(x/2-2)} = \dfrac{1}{x/2+2} = \dfrac{2}{x+4}$

- $I(f) = \int_{x_A=0}^{x_B=4} \dfrac{2}{x+4}\dfrac{\sqrt{5}}{2}dx = \sqrt{5}\int_0^4 \dfrac{1}{x+4}dx = \sqrt{5}\,[\ln(x+4)]_0^4 = \sqrt{5}\,(\ln 8 - \ln 4) = \sqrt{5}\,\ln 2$

IC2

1a) $ds = \sqrt{1+\left(\dfrac{dy}{dx}\right)^2}\,dx \;\rightarrow\; \dfrac{dy}{dx} = \dfrac{2x}{4} - \dfrac{1}{2x} = \dfrac{x^2-1}{2x}$

$\Rightarrow 1+\left(\dfrac{dy}{dx}\right)^2 = 1+\left(\dfrac{x^2-1}{2x}\right)^2 = \dfrac{4x^2+x^4-2x^2+1}{(2x)^2} = \dfrac{x^4+2x^2+1}{(2x)^2} = \dfrac{(x^2+1)^2}{(2x)^2}$

$\Rightarrow ds = \dfrac{x^2+1}{2x}dx = \left(\dfrac{x}{2}+\dfrac{1}{2x}\right)dx = \dfrac{1}{2}\left(x+\dfrac{1}{x}\right)dx$

1b) $l = \int_{x_A=1}^{x_B=e} ds = \int_1^e \dfrac{1}{2}\left(x+\dfrac{1}{x}\right)dx = \dfrac{1}{2}\left[\dfrac{x^2}{2}+\ln x\right]_1^e = \dfrac{1}{2}\left(\dfrac{e^4}{2}+1-\dfrac{1}{2}\right) = \dfrac{1}{4}(e^4+1) \approx 2{,}1\ m$

2) $I_C = \int_1^e \Phi(x,y)\,ds = \int_1^e x\left(\dfrac{x}{2}+\dfrac{1}{2x}\right)dx = \int_1^e \left(\dfrac{x^2}{2}+\dfrac{1}{2}\right)dx = \left[\dfrac{x^3}{6}+\dfrac{x}{2}\right]_1^e = \dfrac{e^3}{6}+\dfrac{e}{2}-\dfrac{1}{6}-\dfrac{1}{2} \approx 4{,}04$

IC3

$y(x) = \sqrt{x}\left(\dfrac{x}{3}-1\right) = x^{1/2}\left(\dfrac{x}{3}-1\right) = \dfrac{x^{3/2}}{3}-x^{1/2} \;\rightarrow\; \Phi(x,y) = xy = x\left(\dfrac{x^{3/2}}{3}-x^{1/2}\right) = \dfrac{x^{5/2}}{3}-x^{3/2}$

1) $ds = \sqrt{1+\left(\dfrac{dy}{dx}\right)^2}\,dx \;\rightarrow\; \dfrac{dy}{dx} = \dfrac{3}{2}\dfrac{x^{1/2}}{3}-\dfrac{1}{2}x^{-1/2} = \dfrac{\sqrt{x}}{2}-\dfrac{1}{2\sqrt{x}}$

$\Rightarrow 1+\left(\dfrac{dy}{dx}\right)^2 = 1+\left(\dfrac{\sqrt{x}}{2}-\dfrac{1}{2\sqrt{x}}\right)^2 = 1+\dfrac{x}{4}-2\dfrac{\sqrt{x}}{2}\dfrac{1}{2\sqrt{x}}+\dfrac{1}{4x} = \dfrac{x}{4}-\dfrac{1}{2}+\dfrac{1}{4x} = \left(\dfrac{\sqrt{x}}{2}+\dfrac{1}{2\sqrt{x}}\right)^2$

$\Rightarrow ds = \left(\dfrac{\sqrt{x}}{2}+\dfrac{1}{2\sqrt{x}}\right)dx = \left(\dfrac{x^{1/2}}{2}+\dfrac{x^{-1/2}}{2}\right)dx$

$l = \int_{x_A=0}^{x_B=3} ds = \int_0^3 \left(\dfrac{x^{1/2}}{2}+\dfrac{x^{-1/2}}{2}\right)dx = \left[\dfrac{2}{3}\dfrac{x^{3/2}}{2}+2\dfrac{x^{1/2}}{2}\right]_0^3 = \dfrac{3^{3/2}}{3}+3^{1/2} = 2\times 3^{1/2} = 2\sqrt{3}$

2)

$I_C = \int_0^3 \Phi(x,y)\,ds = \int_0^3 \left(\dfrac{x^{5/2}}{3}-x^{3/2}\right)\left(\dfrac{x^{1/2}}{2}+\dfrac{x^{-1/2}}{2}\right)dx = \int_0^3 \left(\dfrac{x^3}{6}+\dfrac{x^2}{6}-\dfrac{x^2}{2}-\dfrac{x}{2}\right)dx = \int_0^3 \left(\dfrac{x^3}{6}-\dfrac{x^2}{3}-\dfrac{x}{2}\right)dx$

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$$I_C = [\frac{x^4}{24} - \frac{x^3}{9} - \frac{x^2}{4}]_0^3 = (\frac{81}{24} - \frac{27}{9} - \frac{9}{4}) = \frac{81-72-54}{24} = -\frac{45}{24} = -\frac{15}{8}$$

IC4

- $\begin{cases} x = a\cos t \\ y = a\sin t \\ z = ht \end{cases} \quad t \in [0, 2\pi] \quad \Rightarrow \begin{cases} dx/dt = -a\sin t \\ dy/dt = a\cos t \\ dz/dt = h \end{cases}$
- $ds = \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2 + (\frac{dz}{dt})^2}\, dt = \sqrt{a^2\sin^2 t + a^2\cos^2 t + h^2}\, dt = \sqrt{a^2+h^2}\, dt$
- $L = \int_0^{2\pi} ds = \int_0^{2\pi} \sqrt{a^2+h^2}\, dt = \sqrt{a^2+h^2} \int_0^{2\pi} dt = 2\pi\sqrt{a^2+h^2}$

IC6

- $\rho = a\mathrm{e}^{-\varphi}$, $\varphi \in [0, 2\pi]$
- $ds = \sqrt{(d\rho)^2 + (\rho\, d\varphi)^2} = \sqrt{(\frac{d\rho}{d\varphi})^2 + \rho^2}\; d\varphi = \sqrt{(-a\varphi\mathrm{e}^{-\varphi})^2 + (a\mathrm{e}^{-\varphi})^2}\; d\varphi = \sqrt{2a^2(\mathrm{e}^{-\varphi})^2}\; d\varphi$

  $ds = \sqrt{2}\, a\mathrm{e}^{-\varphi}\, d\varphi$
- $L = \int_0^{2\pi} ds = \int_0^{2\pi} \sqrt{2}\, a\mathrm{e}^{-\varphi}\, d\varphi = a\sqrt{2}[-\mathrm{e}^{-\varphi}]_0^{2\pi} = a\sqrt{2}(-\mathrm{e}^{-2\pi} - (-\mathrm{e}^{-0})) = a\sqrt{2}(1-\mathrm{e}^{-2\pi})$

$\lim_{\varphi \to +\infty} L = a\sqrt{2}(1-\mathrm{e}^{-\infty}) = a\sqrt{2}$

IC9

1) $\Gamma \begin{cases} \rho = R \\ \varphi \in [0, 2\pi] \end{cases} \rightarrow \begin{cases} x = R\cos\varphi \\ y = R\sin\varphi \end{cases}$, R = cte

- $\lambda(x, y) = C$, C = cte

$Q = \oint_\Gamma C\, dl = C \oint_\Gamma dl = 2\pi R C$, car $\oint_\Gamma dl$ = périmètre du cercle

- $\lambda(x, y) = \frac{Cx^2}{x^2+y^2}$ → coordonnées polaires ⇒ $\lambda(R, \varphi) = \frac{C R^2\cos^2\varphi}{R^2\cos^2\varphi + R^2\sin^2\varphi} = C\cos^2\varphi$

$dl = \sqrt{(d\rho)^2 + (\rho\, d\varphi)^2} = \sqrt{(dR)^2 + (R d\varphi)^2} = R\, d\varphi$

$Q = \oint_0^{2\pi} C\cos^2(\varphi) R\, d\varphi = RC \oint_0^{2\pi} \frac{1+\cos(2\varphi)}{2} d\varphi = \frac{RC}{2}[\varphi + \sin(2\varphi)]_0^{2\pi} = \pi RC$

2) $\Sigma \begin{cases} \rho \in [0, R] \\ \varphi \in [0, 2\pi] \end{cases} \rightarrow \begin{cases} x = \rho\cos\varphi \\ y = \rho\sin\varphi \end{cases}$

- $\sigma(x, y) = C$, C = cte

$Q = \iint_\Sigma C\, dS = C \iint_\Sigma dS = \pi R^2 C$, car $\iint_\Sigma dS$ = aire du disque

- $\sigma(x, y) = C\sqrt{x^2+y^2}$ → coordonnées polaires ⇒ $\sigma(\rho, \varphi) = C\rho$

$dS = \rho\, d\rho\, d\varphi$

$Q = \int_0^{2\pi} d\varphi \int_0^R C\rho^2\, d\rho = 2\pi C \frac{R^3}{3}$

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3) E $\begin{cases} r \in [0, R] \\ \theta \in [0, \pi] \\ \varphi \in [0, 2\pi] \end{cases} \rightarrow \begin{cases} x = r \sin\theta \cos\varphi \\ y = r \sin\theta \sin\varphi \\ z = r \cos\theta \end{cases}$

- $\rho(x, y, z) = C$, C = cte

$Q = \iiint_E C \, dV = C \iiint_E dV = \frac{4}{3} \pi R^3 C$, car $\iint_\Sigma dS$ = volume de la sphère

- $\rho(x, y, z) = C\sqrt{x^2 + y^2 + z^2}$ → coordonnées sphérq ⇒ $\rho(r, \theta, \varphi) = Cr$

$dV = r^2 \sin\theta \, dr \, d\theta \, d\varphi$

$Q = \int_0^{2\pi} d\varphi \int_0^{\pi} \sin\theta \, d\theta \int_0^{R} C r^3 dr = 2\pi 2C \frac{R^4}{4} = \pi C R^4$

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Notes :
cte : constante(s)
-q : -ique(s)

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